ChatGPT

Polymetallic Nodule Potential around Falklands and South Sandwich Islands

Executive Summary: Polymetallic nodules are ferromanganese concretions found on deep, flat abyssal plains【81†L160-L168】. They grow layer-by-layer from trace metals delivered by sinking plankton under conditions of *extremely* low sedimentation【81†L167-L175】【85†L211-L219】. Globally, major nodule provinces occur on stable old seafloor (e.g. Pacific Clarion–Clipperton and Peru Basins) 【85†L198-L202】【81†L160-L168】. In contrast, the South Atlantic near the Falklands/South Sandwich has no mapped fields. Candidate areas must be inferred by analogy to known provinces. Of these, the deep Argentine Basin (east of Falklands, depths ~4–5 km) most closely meets the depth and low-sediment criteria【85†L211-L219】【81†L160-L168】. By contrast, the Falkland Plateau (bathymetry <2.5 km【55†L108-L116】) and the tectonically active Scotia–Sandwich arc (steep slopes and volcanism 【55†L139-L144】) are geologically unfavorable. We estimate **low** likelihood of high nodule abundance on the Falkland Plateau (due to shallow depth and high terrigenous input) but **medium** potential in portions of the Argentine Basin (suitable depth, fine sediment). Figure 1 (below) shows a typical nodule sample for context. Our regional assessment (next) ranks each subregion by its nodule-forming attributes and confidence level.

【95†embed_image】 *Figure 1: Example ferromanganese nodule (manganese-rich concretion) from ~4000 m depth. Such nodules accrete slowly on flat deep seafloors under very low sedimentation【81†L167-L175】 【85†L211-L219】.*

Detailed Regional Assessment

Falkland Plateau and Shelf (< ~2.5 km)

The Falkland Islands lie on a broad continental block (Falkland Plateau) extending east from South America【55†L108-L116】. Water depths here are relatively shallow (<2.5 km) and sediment accumulation is thick (glacial and shelf detritus). Strong currents (Malvinas/Falkland Current) and biological productivity flank the islands, but high terrigenous clay input tends to bury nascent nodules. Because nodules require *very* low sedimentation【85†L211-L219】, the high depositional regime of this shelf–basin argues **against** significant nodule fields. In effect this area resembles shelf-edge basins with high accumulation rather than abyssal plains【81†L167-L175】【85†L211-L219】. We assign **low confidence** (low likelihood) for nodules on the Falkland Plateau.

Argentine Basin (East of Falklands; ~3–5 km depth)

East of the plateau lies the Argentine Basin (also called Falkland Trough), a deep abyssal plain reaching ~4–5 km. This open-ocean basin is under the influence of the Antarctic Circumpolar Current, which winnows sediments. The seafloor substrate here is mainly pelagic clays with low terrigenous supply. Sedimentation rates in comparable Southwest Atlantic basins are expected to be very low ($\lesssim$5 mm/kyr) 【85†L211-L219】, meeting the key nodule criterion of “extremely low sedimentation.” Depths of ~4–5 km also fall into the known nodule range【85†L211-L219】. Productivity proxies (e.g. surface chlorophyll, export flux) may be moderate near the Subantarctic front, supplying Mn/Fe to the bottom. Geologically this oceanic crust is old and quiescent. In sum, the Argentine Basin has the most favorable

combination of flat abyssal bathymetry, slow deposition and metal flux (analogous to the Pacific Peru Basin). We judge a **medium confidence** of nodule occurrence here, subject to confirmation by survey.

Scotia Sea and South Sandwich (Volcanic Arcs)

To the south and east, the Scotia Sea is bounded by the North and South Scotia Ridges, including South Georgia and the South Sandwich volcanic chain【55†L108-L116】【55†L139-L144】. Although some basins in the Scotia Sea reach abyssal depths (~3–4 km), the region is geologically complex (active arc volcanism, trenches). The South Sandwich Islands (a volcanic arc) descend into a trench, and the local seafloor is largely volcanic basement and carbonate turbidites. In general, tectonic activity and rough topography inhibit extensive flat plains. Currents are vigorous (ACC) and productivity is seasonally high, but sediments here also reflect dynamic glacial currents and turbidites. Given the principles from Smithsonfian et al. – nodules **do not** form in tectonically active or sediment-rich settings【81†L172-L175】 – the Scotia/Sandwich sector is **very low confidence**. Only limited patches of abyssal plain (if any) exist, and these are small relative to Pacific provinces.

Site Confidence: We therefore focus on one prime nodule target: the deep Argentine Basin (moderate potential). The Falkland Plateau region is unlikely (shallow/sediment), and the Scotia–Sandwich area is far less promising (volcanic and rough). Table 1 (below) compares these candidates by depth, substrate, sedimentation and productivity.

Candidate Site Comparison

Area	Depth (m)	Seabed/ Substrate	Sedimentation	Productivity Indicators	Nodule Likelihood
Falkland Plateau	~0–2500	Glacial/ terrigenous clay	**High**	High (Falkland Current, shelf)	**Low**
Argentine Basin	~3500–5000	Pelagic clay on abyssal plain	**Low–Moderate**	Moderate (sub-Antarctic waters)	**Medium**
South Sandwich Arc	~3000–6000	Volcanic basement & ooze	**Moderate**	Low–Moderate (ACC influence)	**Very Low**

Table 1. Comparison of candidate regions. Productivity proxies include e.g. surface chlorophyll and dust flux. Likelihood is qualitative ("Low/Medium/High") for nodules.

Recommended Data Sources

- **Global Bathymetry:** Nippon/GEBCO 15″ grid (GEBCO_2023) provides global depths【10†L71-L79】. Also EMODnet Bathymetry (European waters), NOAA ETOPO1, and IBCSO (for Southern Ocean).
- **Sediment and Core Data:** NOAA NCEI sediment thickness maps; PANGAEA and IODP databases for sediment core records (accumulation rates, lithology) around the South Atlantic. SCAR/ACE core data for Southern Ocean proxies.
- **Geochemical Surveys:** USGS/ODP published geochemical analyses of Atlantic nodules and sediments; literature (e.g. Kasten et al.) on South Atlantic rare-earth content.
- **Currents & Productivity:** Argo/GO-SHIP profiles and satellite data (NASA MODIS chlorophyll, WINDSAT iron flux) to infer organic flux. Ocean reanalysis (HYCOM, ECCO) for current patterns.

- **Tectonics/Substrate:** Global seafloor age grids (e.g. Müller et al.) and magnetic anomaly maps to estimate crust age (older = more stable). Marine seismic reflection profiles (BAS/USGS data) to assess sediment thickness and substrate type.

Each dataset should be examined to refine bathymetry, sedimentation rates, and chemistry. For example, GEBCO/EMODnet bathymetry will locate flat abyssal plain areas; sediment cores and radiometric dating can quantify sedimentation (mm/kyr); biogeochemical tracers (Ba, Ti) can estimate productivity; seismic lines can reveal crustal relief and underlying sediment.

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flowchart LR
    A[Define region of interest] --> B[Acquire bathymetry (GEBCO/EMODnet)]
    B --> C[Map physiographic features (abyssal plains, plateaus, trenches)]
    C --> D[Compile sedimentation data (cores, proxies)]
    C --> E[Identify analog nodule provinces]
    C --> F[Assess oceanography (currents, productivity proxies)]
    D --> G[Evaluate sedimentation rates & thickness]
    E --> G
    F --> G
    G --> H[Score candidate areas on nodule criteria]
    H --> I[Produce maps, tables, and report]
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Figure 2. Workflow of the nodule potential analysis.

Sources: Literature on seafloor nodules and Southern Ocean geology【81†L160-L168】【55†L108-L116】【85†L211-L219】 and global marine datasets were used to construct this assessment. The key references and datasets listed above should be consulted for detailed data.